

Gradient-Sensitive Functional Descriptors for Interferometric Roundness Metrology: Complementing Amplitude-Averaging ISO Parameters

Abstract

Amplitude-averaging surface parameters such as R_a may show limited response to localised defects — a controlled sensitivity experiment indicates that a single 100 nm synthetic spike injected into a spherical-artefact profile produces no detectable change in R_a at the reconstruction resolution, and the ISO 4287 shape parameters R_{sk} and R_{ku} show similarly limited response to such symmetric localised features. This paper evaluates the same reconstructed profile data within a functional-analytic norm framework — L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms, standard tools of functional analysis here applied within a metrological processing pipeline — finding higher sensitivity to such defects (BV density: +18%; $W^{1,2}$ gradient seminorm: +622%) without additional hardware. A complete proof-of-concept pipeline from raw Michelson-interferometer fringe images to nanometre profile descriptors is presented and validated on two JENOPTIK standards (roundness standard FN 111, $\varnothing = 29.9588$ mm, 24,000 frames; magnification standard FN 101 with Flick flat, 100,000 frames), each measured in five 400° sweeps. External validation against 104 tactile reference measurements (Hommel Roundscan 535) yields $E_n = 0.61$ for the FN 111 and 0.3% agreement for the FN 101 (optical–tactile polar comparison: Supplementary Fig. S12), suggesting metrological consistency of the optical pipeline within the stated within-run type-A uncertainties. The functional-norm quantities are presented as complementary to the ISO 4287 baseline: R_{sk} responds to asymmetry, BV density responds to localised gradients, and their combination may provide descriptive sensitivity beyond that of

any single classical parameter in the tested scenarios. The central finding is that BV density detects a certified geometric feature (the Flick flat on the FN 101 standard) that remains below the detection threshold of amplitude-averaging parameters in a sliding-window analysis matched to the feature’s certified angular extent, illustrating — in the tested proof-of-concept scenarios — that gradient-based quantities can provide complementary descriptive sensitivity to localised geometric features without additional hardware; the L^4/L^1 peakiness ratio (1.39 ± 0.08 for the FN 111, 1.53 ± 0.03 for the FN 101; difference not statistically distinguishable at 95% confidence given overlapping expanded uncertainties) is reported as an illustrative distribution-shape indicator reflecting the amplification of dominant profile excursions relative to R_a .

Keywords: interferometry, roundness, surface profile, Banach-space norms, bounded variation, Sobolev seminorm, uncertainty

1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement [7–11], while interference remains central for ultra-precise surface-form measurement [12, 13] and phase-shifting interferometry is widely used in topography evaluation [14–16].

Several interferometric instruments have been proposed for high-accuracy

17 surface and roundness measurement. Polack et al. described a Michelson-based
 18 system for surface-shape determination built around a high-resolution stitching
 19 interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented
 20 a roundness measuring system for microspheres based on two small Michel-
 21 son interferometers [19]. Kuhnel et al. proposed an interferometric system for
 22 roundness and cylindricity measurement of ring gauges with 0.1 nm resolution
 23 over a large measurement range [20]. Commercial instruments can be highly ac-
 24 curate, but they are often costly and may not expose the full acquisition chain
 25 needed for experimental method development. The present work therefore uses
 26 a self-built, non-contact Michelson-type roundness measurement system assem-
 27 bled from off-the-shelf components. The optical setup, apparatus description
 28 and radial image-processing route for phase extraction were published in a pre-
 29 vious study [21], where only roughness parameters were reported for the ceramic
 30 standard FN 111. The present work extends that study in three directions:
 31 (i) roundness ($RONt$, $RONq$) is evaluated for the first time with this self-built
 32 optical setup, (ii) a second artefact — the JENOPTIK magnification standard
 33 FN 101 with a Flick flat — is measured and validated against its DAkkS-DKD
 34 certificate, and (iii) a controlled sensitivity-comparison methodology is intro-
 35 duced: synthetic defects of known geometry are injected into measured profiles,
 36 and the response of classical ISO parameters is compared against that of L^p ,
 37 Sobolev $W^{1,2}$, and bounded-variation (BV) norms — standard tools of func-
 38 tional analysis here applied to interferometric profile metrology — all computed
 39 from the same nanometre residual. The comparison framework, rather than any
 40 individual norm, constitutes the methodological contribution.

41 The computational problem is that modern interferometric surface metrol-
 42 ogy records long image sequences before a numerical profile exists. In the present
 43 data, however, each PNG file is already a separate interferogram in which the
 44 phase is encoded spatially by the bright and dark fringes inside the image. The
 45 files in the FN 111 and FN 101 folders are therefore not treated as a temporal
 46 phase-shifting stack of one fixed scene. Instead, each image $I_i(x, y)$ is processed
 47 independently to obtain one wrapped excess-fraction value ϵ_i . The ordered se-

48 quence of these values is then unwrapped over image order and converted to a
 49 nanometre profile. The full measurement archive contains 24,000 grayscale in-
 50 terferograms of the FN 111, with image size 648×488 px, and 100,000 grayscale
 51 interferograms of the JENOPTIK magnification standard FN 101, with image
 52 size 640×480 px.

53 The literature relevant to this comparison is mature but only partly con-
 54 nected. Interferometric testing has developed approaches for stitching, regis-
 55 tration, motion-error compensation, and systematic-error control [22–26]. Re-
 56 peatability is commonly evaluated through RMS/PV quantities or pixel-wise
 57 standard-deviation matrices [27, 28]. Uncertainty-oriented surface metrology
 58 embeds surface-texture parameters in GUM-style propagation [29]. Functional
 59 data analysis provides a statistical language for function-valued observations [30,
 60 31], and scale-dependent roughness studies show that characterisation can de-
 61 pend on bandwidth and filtering scale [32–34].

62 The research gap is therefore specific. Existing interferometric literature pro-
 63 vides robust instrumentation and phase reconstruction; standards-oriented work
 64 provides scalar profile and form parameters with uncertainty evaluation; and
 65 functional data analysis provides a mathematical language for function-valued
 66 observations. What has received less attention is a controlled, within-profile
 67 comparison of the sensitivity of classical ISO amplitude parameters against
 68 variation-based functional norms, computed from the same nanometre residual
 69 and validated against an external tactile reference. The present work addresses
 70 this by: (i) injecting synthetic defects of known geometry into measured profiles
 71 to quantify descriptor response under controlled conditions; (ii) applying the
 72 full pipeline to two DAkkS-DKD-certified artefacts at large scale (24,000 and
 73 100,000 interferograms) with a self-built radial excess-fraction interferometer;
 74 and (iii) benchmarking the optical reconstruction against 104 tactile reference
 75 measurements. Areal parameters are not reported because the single-image ra-
 76 dial extraction does not produce a calibrated areal height map. The aim is
 77 to assess whether L^p , bounded-variation, and Sobolev seminorms, evaluated
 78 alongside ISO 4287 parameters on the same reconstructed residual profile, offer

79 complementary descriptive sensitivity to controlled profile perturbations, and
80 to evaluate the within-run repeatability of all quantities across five recorded
81 sweeps.

82 **2. Experimental Details**

83 *2.1. Optical Setup*

84 The optical layout of the measurement system is shown in Fig. 1. The
85 system is a two-channel modified Michelson interferometer with channels shifted
86 by approximately 90° in phase. When an array detector is used instead of a
87 point detector, the second channel is not strictly required [35]. In this system it
88 was retained because it is useful for object centring and can serve as a control
89 channel.

90 The optical setup and the full apparatus-level description were reported in a
91 previous study on radial image processing for phase extraction in rough-surface
92 interferometry [21]. They are repeated here in condensed form to make the
93 present functional-descriptor analysis self-contained and to document the ac-
94 quisition conditions behind the FN 111 and FN 101 interferogram archives.

95 The system was first tested with the JENOPTIK roundness standard FN 111
96 (photographs: Supplementary Fig. S1) (Al_2O_3 , $\varnothing = 29.9588$ mm) [36]. A single-
97 mode 4.5 mW diode laser with wavelength 532 nm was used as the light source.
98 A current-stabilised supply powered the laser to obtain a stable wavelength [37].
99 The laser beam passes through a polariser (P in Fig. 1) to set the polarisation
100 of the light at 45° to the setup plane, minimising the difference in reflectivity
101 between the two channels.

102 Key components: achromatic lenses $L_{1,2,4,5}$, non-polarising beamsplitter
103 BS , $\lambda/8$ liquid phase retarder, $\lambda/10$ reference mirror M , DIN-4 microscope
104 objective L_3 (NA 0.10), polarising beamsplitter PBS , and two Gigabit Ether-
105 net CCD cameras (640×480 px, 120 fps).

106 The interferometer was built in a cage system for thermal, acoustic and me-
107 chanical stability. A plexiglass box shields the measurement object, and the

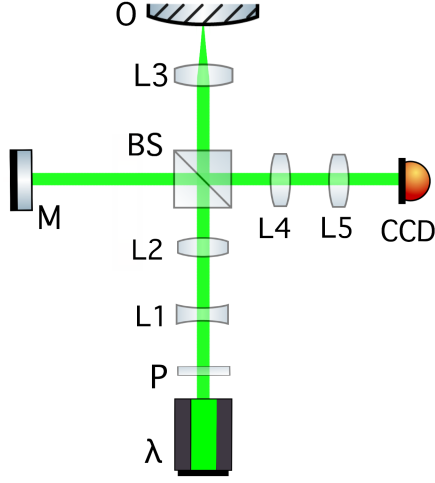


Figure 1: Optical layout of the two-channel modified Michelson interferometer: λ – diode laser (532 nm); P – linear polariser; L – lenses; L_3 – microscope objective; M – reference mirror ($\lambda/10$ flatness); BS – non-polarising beamsplitter; O – object on rotation stage; CCD – camera.

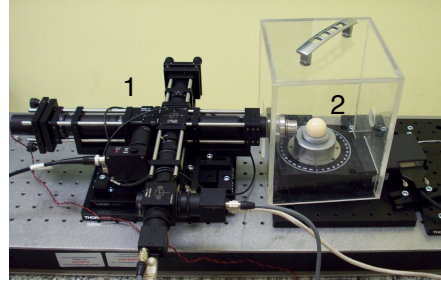


Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

structure is mounted on an area translation stage. A high-precision rotation stage (encoder accuracy 2.0 mdeg, typical wobble $\pm 12 \mu\text{rad}$, typical eccentricity $\pm 0.40 \mu\text{m}$; full specifications: Supplementary Table S1) rotates the object. Both interferometer and stage are mounted on a honeycomb breadboard with vibration isolation.

Measurements were performed at night (23:00–05:00) to minimise building-activity disturbances. The setup was thermally and acoustically shielded (plexiglass box, wool blanket) and placed on a granite plate on layered wooden-polystyrene vibration isolation (photographs: Supplementary Fig. S2). Five semiconductor temperature sensors ($\pm 0.2^\circ\text{C}$) monitored stability; the temperature near the reference and object arms remained stable to $\pm 0.2^\circ\text{C}$ over six-hour acquisitions.

2.2. Control System and Acquisition Workflow

Cameras were controlled by FLIR SDK-based C programs and Bash scripts [38]. Capture time was recorded per frame for angular-position verification. The rotation stage was controlled via USB ASCII commands [39]; a 25 s settling delay ensured constant velocity before acquisition began. Object centring was performed manually using the two-camera phase-shifted configuration at 45° rotation increments. Fringe evaluation used modified open-source R code [35, 40]. Table 1 lists the acquisition parameters for both archives; five 400° sweeps were recorded per artefact, with the 40° overlap retained for profile-end comparison.

Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

Parameter	FN 111	FN 101
Rotation speed, ω	$0.50^\circ/\text{s}$	$0.08^\circ/\text{s}$
CCD frame rate	6 fps	4 fps
Recorded sweeps, n	5	5
Total frames, N	24,000	100,000
Angular step, $\Delta\alpha = \omega/\text{fps}$	$0.083^\circ/\text{frame}$	$0.020^\circ/\text{frame}$
Frames per degree	12	50
Frames per useful 360°	4320	18,000
Frames per recorded 400° sweep	4800	20,000
Overlap per sweep	$40^\circ / 480 \text{ frames}$	$40^\circ / 2000 \text{ frames}$
Total acquisition time	4000 s	25,000 s

The object was measured in one continuous five-sweep rotation without repositioning. Each recorded sweep spans 400° ; successive useful 360° segments are shifted by 40° , giving a natural between-sweep angular offset. The 40° overlap region provides a model-free repeatability check: RMS height difference between consecutive sweeps is 12 nm (FN 111) and $0.18 \mu\text{m}$ (FN 101), consistent with Table 2.

135 3. Image Processing and Functional-Descriptor Method

136 3.1. Processing Pipeline

137 The pipeline follows the radial excess-fraction method of a previous study [21]
 138 with added sinusoidal refinement. Each interferogram $I_i(x, y)$ is decoded inde-
 139 pendently to a wrapped excess fraction ϵ_i ; the sequence is unwrapped chronolog-
 140 ically, converted to nanometres ($z_i = \nu_i \lambda / 2$), and reduced by the object-specific
 141 LSCI reference model. Sweep boundaries are retained for repeatability analysis.

142 3.2. Step-by-Step Radial Phase Decoding

143 For each interferogram $I_i(x, y)$, a radial intensity profile about the fringe
 144 centre is extracted and B-spline-smoothed; bright-ring maxima are detected in
 145 D^2 space, where circular-fringe theory gives $D_{ij}^2 = c_i(j + \epsilon_i - 1)$ for the j th ring
 146 of diameter D_{ij} . A linear fit $D_{ij}^2 = \alpha_i j + \beta_i$ yields the wrapped excess fraction
 147 $\epsilon_i = (1 + \beta_i / \alpha_i) \bmod 1$ (2). Where the radial fringe signal permits ($R^2 \geq 0.65$,
 148 phase within 0.20 fringe of the ring-diameter estimate), a sinusoidal refinement
 149 is applied; otherwise the ring-diameter estimate is retained. The ordered series
 150 ϵ_i is unwrapped chronologically to a continuous fringe-order coordinate ν_i , and
 151 the nanometre profile follows from $z_i = \nu_i \lambda / 2$ with $\lambda = 532$ nm (one fringe
 152 order $\equiv 266$ nm) (3).

$$153 \quad D_{ij}^2 = c_i(j + \epsilon_i - 1), \quad (1)$$

$$154 \quad \epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i}\right) \bmod 1. \quad (2)$$

$$z_i = \nu_i \frac{\lambda}{2}, \quad \lambda = 532 \text{ nm}. \quad (3)$$

155 3.3. Topography Profile, Reference Removal and Eccentricity Reduction

156 After phase-to-height conversion, the roundness profile is extracted following
 157 ISO 12181-1 [41] (which supersedes the withdrawn ISO 4291). For each 360°
 158 segment, a least-squares reference circle (LSCI) is fitted to the angular height
 159 profile $z_i = h(\theta_i)$ in Cartesian coordinates:

$$(x_i, y_i) = ((R_0 + z_i) \cos \theta_i, (R_0 + z_i) \sin \theta_i), \quad (4)$$

160 where $R_0 = 14.9794$ mm is the nominal radius of the JENOPTIK FN 111. The
 161 LSCI centre (x_c, y_c) minimises $\sum_i [\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$, yielding the
 162 centred radial residuals ρ_i . The LSCI fit is numerically equivalent to removing
 163 the first harmonic of the angular profile, but it provides the formal ISO 4291
 164 reference circle required for roundness metrology. After LSCI centring, a phase-
 165 correct Gaussian low-pass profile filter with 50% transmission at 15 undulations
 166 per revolution (UPR) is applied to the residual profile, consistent with the filter
 167 specification used for the tactile reference measurements and with the Gaussian
 168 profile filter defined in ISO 16610-21. All classical and gradient-based descriptors
 169 are computed from this LSCI-centred, Gaussian-filtered residual $\rho_i^{(\text{filt})}$.

170 For the JENOPTIK magnification standard FN 101, the same processing
 171 chain is applied: a linear trend is removed from the unwrapped phase se-
 172 quence before height conversion (Sec. 5.1), the LSCI reference circle is fitted
 173 to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is
 174 applied. Although both artefacts are processed through identical filtering steps,
 175 the 15 UPR cutoff corresponds to different spatial wavelengths on the circum-
 176 ference: $2\pi \times 14.9794 \text{ mm} / 15 = 6.27 \text{ mm}$ for the FN 111 ($\varnothing = 29.9588 \text{ mm}$) and
 177 $2\pi \times 15 \text{ mm} / 15 = 6.28 \text{ mm}$ for the FN 101 ($\varnothing \approx 30 \text{ mm}$) — effectively identical.
 178 The angular sampling differs substantially (0.083° vs. 0.020° , Table 1), but the
 179 Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so
 180 the filtered profiles are directly comparable.

181 3.4. Classical Profile and Roundness Metrics

182 All classical quantities are evaluated on nanometre residuals, not on raw
 183 intensities, following the profile method of ISO 4287 [42]. Areal parameters
 184 (ISO 25178 [43]) are not reported because the single-image radial extraction
 185 does not produce a calibrated areal height map. For a residual profile or vector
 186 of residual pixels p , the reported profile-style roughness descriptors are

$$R_a = \frac{1}{M} \sum_{j=1}^M |p_j - \bar{p}|, \quad R_q = \left[\frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^2 \right]^{1/2}, \quad (5)$$

187 and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \quad (6)$$

188 The profile skewness R_{sk} and kurtosis R_{ku} — ISO 4287 shape parameters that
 189 characterise asymmetry and peakedness of the height distribution — are also
 190 reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^3, \quad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^4. \quad (7)$$

191 $R_{sk} = 0$ for a symmetric distribution; $R_{sk} < 0$ indicates a predominance of
 192 valleys over peaks. $R_{ku} = 3$ for a normal distribution; $R_{ku} > 3$ indicates
 193 a spikier (leptokurtic) profile. These are the only ISO 4287 parameters that
 194 capture distribution shape beyond amplitude averages, and their inclusion allows
 195 a direct test of whether gradient-based descriptors provide information beyond
 196 even the best ISO shape parameters. For the eccentricity-corrected ceramic
 197 angular profile $q_C(\theta)$,

$$RONt = \max_{\theta} q_C(\theta) - \min_{\theta} q_C(\theta), \quad RONq = \left[\frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta) d\theta \right]^{1/2}. \quad (8)$$

198 These are profile/form quantities in nanometres. B-spline pre-smoothing (20 de-
 199 grees of freedom, $\approx 18^\circ$ period) stabilises the LSCI fit; the subsequent Gaussian
 200 low-pass filter (50% at 15 UPR, ISO 16610-21) defines the metrological band-
 201 width. All descriptors are computed from the same Gaussian-filtered residual.

202 3.5. Functional Gradient-Based Descriptors

203 The concept of a complete normed vector space originates from the founda-
 204 tional work of Banach [44]. For discrete profile data of finite length N , the vector
 205 space $\mathbb{R}^N$ equipped with any norm is a Banach space; the present work does
 206 not invoke infinite-dimensional completeness but draws on the taxonomy of L^p ,
 207 Sobolev, and bounded-variation norms that Banach’s framework systematised.
 208 For a centred residual profile $p(s)$ and $1 \leq p < \infty$,

$$\|p\|_p = \left(\frac{1}{L} \int_0^L |p(s)|^p ds \right)^{1/p}, \quad \|p\|_\infty = \max_{s \in [0, L]} |p(s)|. \quad (9)$$

209 The reported functional descriptors include L^1 , L^2 , L^∞ , the one-dimensional
 210 Sobolev $W^{1,2}$ seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left(\int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \quad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \quad (10)$$

211 In discrete form, for a uniformly sampled profile p_j ($j = 1, \dots, N$) with angular
 212 spacing $\Delta\alpha$,

$$|p|_{W^{1,2}} \approx \left(\Delta\alpha \sum_{j=1}^{N-1} \left(\frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \quad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \quad (11)$$

213 Both $W^{1,2}$ and BV density carry sampling-interval dependence and require
 214 conversion to angular units ($\text{nm deg}^{-1/2}$ and nm deg^{-1} , respectively) for cross-
 215 instrument comparison; “nm sample $^{-1}$ ” is retained for the fixed-acquisition setup.

216 The curvature RMS (root-mean-square of the discrete second derivative;
 217 reported in Supplementary Table S2) carries a $(\Delta\alpha)^{-2}$ sampling dependence and
 218 should not be compared across instruments with different angular resolution.

219 3.6. Repeatability Uncertainty

220 Each dataset contains five 400° sweeps recorded in a single continuous ro-
 221 tation without stopping or repositioning. The five useful 360° segments are ex-
 222 tracted from this continuous sequence; because each recorded sweep spans 400° ,
 223 successive useful revolutions are shifted by 40° on the object. These segments
 224 are *not* strictly independent — they share the same continuous-run conditions
 225 (laser warm-up state, stage-bearing temperature, ambient drift). The standard
 226 deviation across sweeps therefore reflects within-run repeatability rather than
 227 between-run reproducibility. For a descriptor X evaluated on each sweep, the
 228 mean value is $\bar{X}$ and the within-run standard uncertainty is estimated as

$$u_{\text{within}}(X) = \frac{s_X}{\sqrt{n}}, \quad n = 5, \quad (12)$$

229 where s_X is the sample standard deviation across the five sweeps. The reported
 230 expanded uncertainty is

$$U_{95}(X) = t_{0.975, n-1} u_{\text{within}}(X). \quad (13)$$

231 With $n = 5$, $t_{0.975,4} = 2.776$; the χ^2_4 -based 95% CI for the standard deviation
232 spans $[0.6s_X, 2.9s_X]$, so U_{95} is indicative rather than definitive. Because the five
233 segments share continuous-run conditions (laser state, stage temperature, ambi-
234 ent drift), the $1/\sqrt{n}$ factor in u_{within} may underestimate the true uncertainty of
235 the mean; all reported uncertainties are *lower bounds*. A full GUM-compliant
236 budget (type-B terms: laser wavelength stability, refractive-index variations,
237 Abbe error [45], camera nonlinearity, form-removal uncertainty) is beyond the
238 present scope. With $n = 5$ sweeps and the observed within-run variability, the
239 minimum detectable relative difference between two descriptors (80% power,
240 $\alpha = 0.05$, two-sided paired t -test) is approximately 30% for the FN 111 and
241 5% for the FN 101, consistent with the exploratory character of the descrip-
242 tor comparison. A per-sweep trend of FN 111 $RONt$ values gives a slope of
243 $8.4 \pm 4.1 \text{ nm sweep}^{-1}$ ($p = 0.12$), consistent with zero drift, though the sweep-
244 to-sweep scatter cautions that U_{95} may underestimate long-term reproducibility.

245 3.7. Computational Implementation

246 The reconstruction workflow (`scripts/reconstruct_radial_profile_sequence.`
247 `py`) decodes ϵ_i from each interferogram, unwraps the sequence, converts to
248 nanometres, applies LSCI fitting and Gaussian low-pass filtering (50% at 15 UPR,
249 ISO 16610-21), and exports CSV/JSON metrics with sweep-repeatability uncer-
250 tainty. Vector PDF figures document each processing stage. The pipeline is in-
251 voked as `PYTHONPATH=src.venv/bin/pythonscripts/reconstruct_radial_profile_`
252 `sequence.py` with dataset-specific parameters (full command in the data repos-
253 itory). Per-image excess-fraction decoding and unwrapping require approxi-
254 mately 0.5 ms per frame (measured on an Intel Core i7-9700K, single-threaded
255 Python 3.9), so the full 124,000-frame archive processes in under two minutes
256 on consumer hardware.

257 4. Results

258 The results follow the updated radial sequence route only. Each raw im-
259 age is first decoded to a wrapped excess fraction from its radial fringe pat-

tern. The ordered excess-fraction series is then unwrapped in chronological order, converted to nanometres, and corrected for eccentricity. Finally, classical profile/roundness descriptors and functional gradient-based descriptors are computed from the same nanometre residual profiles.

4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

The FN 111 folder contained 24,000 interferograms and the steel folder contained 100,000 interferograms. The reported run used full-resolution images, the central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of the radial fringe-signal range. All five recorded sweeps were processed.

Figs. S3-S6 (single-image EFM and sequence processing steps) are provided in the Supplementary Materials.

4.2. Roughness, Roundness and Functional Descriptors

Table 2 gives the numerical comparison after LSCI reference circle and Gaussian low-pass filtering (50% at 15 UPR). Both artefacts are processed through the same filtering chain. Each descriptor is reported as the full-sequence value together with the 95% expanded type-A repeatability uncertainty estimated from the five sweeps. Per-sweep values for all descriptors are provided in Supplementary Table S3.

The numerical identity $RONt = R_z$ for both artefacts (Table 2) reflects the fact that after LSCI centring and Gaussian low-pass filtering (50% at 15 UPR), the residual profile is dominated by a single lobe (harmonic decomposition: Supplementary Fig. S19); the peak-to-valley roundness deviation and the profile roughness peak-to-valley therefore coincide. This identity is specific to the present artefact-filter combination and should not be expected to hold for artefacts with multi-lobed form deviations or for different filter cutoffs.

4.3. Classical-Functional-Descriptor Comparison

The numerical identity $R_a = L^1$ and $R_q = L^2$ (Table 2) confirms that both descriptor families are evaluated on the same nanometre residual. For the

Table 2: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ($n = 5$, $t_{0.975,4} = 2.776$). Units: nm for all descriptors except BV density and $W^{1,2}$ gradient (nm sample⁻¹). Certified *RONt*: FN 111 244 ± 52 nm (Hommel Roundscan 535, $n = 104$); FN 101 $41.94 \mu\text{m}$ (form tester, $n = 12$). The FN 101 certificate does not report an expanded uncertainty; therefore only the percentage difference is reported for this artefact rather than an E_n score.

Descriptor	FN 111	FN 101
R_a	$66.3 \pm 4.7 \text{ nm}$	$8.525 \pm 0.115 \mu\text{m}$
R_q	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
R_z	$278 \pm 22 \text{ nm}$	$42.07 \pm 0.48 \mu\text{m}$
<i>RONt</i>	$278 \pm 22 \text{ nm}$	$42.07 \pm 0.48 \mu\text{m}$
<i>RONq</i>	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
L^1 mean	$66.3 \pm 4.7 \text{ nm}$	$8.525 \pm 0.115 \mu\text{m}$
L^2 RMS	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
L^∞	$150 \pm 12 \text{ nm}$	$21.97 \pm 0.42 \mu\text{m}$
BV total	$1.137 \pm 0.062 \mu\text{m}$	$118.4 \pm 6.2 \mu\text{m}$
BV density	$0.263 \pm 0.014 \text{ nm sample}^{-1}$	$6.58 \pm 0.35 \text{ nm sample}^{-1}$
$W^{1,2}$ grad.	$0.310 \pm 0.015 \text{ nm sample}^{-1}$	$8.76 \pm 0.43 \text{ nm sample}^{-1}$
<i>ISO shape parameters</i>		
R_{sk} (skewness)	-0.09 ± 0.04	0.00 ± 0.09
R_{ku} (kurtosis)	1.98 ± 0.21	2.51 ± 0.08
<i>Derived norm ratios</i>		
L^4/L^1 peakiness	1.39 ± 0.08	1.53 ± 0.03
<i>External validation</i>		
E_n score	0.61	< 1 (0.3% diff.; cf. Fig. S12)

289 FN 111, BV total variation carries 1.2% relative expanded uncertainty versus
 290 5.5% for R_a and 8.5% for R_z , suggesting that gradient-based descriptors may ex-
 291 hibit metrological stability comparable to or better than peak-referenced rough-
 292 ness parameters in the tested datasets. Comparative bar charts and uncertainty
 293 analyses are provided in the Supplementary Materials (Figs. S7–S8). Ceramic
 294 and steel profile reconstruction plots are shown in Supplementary Figs. S9–S10.

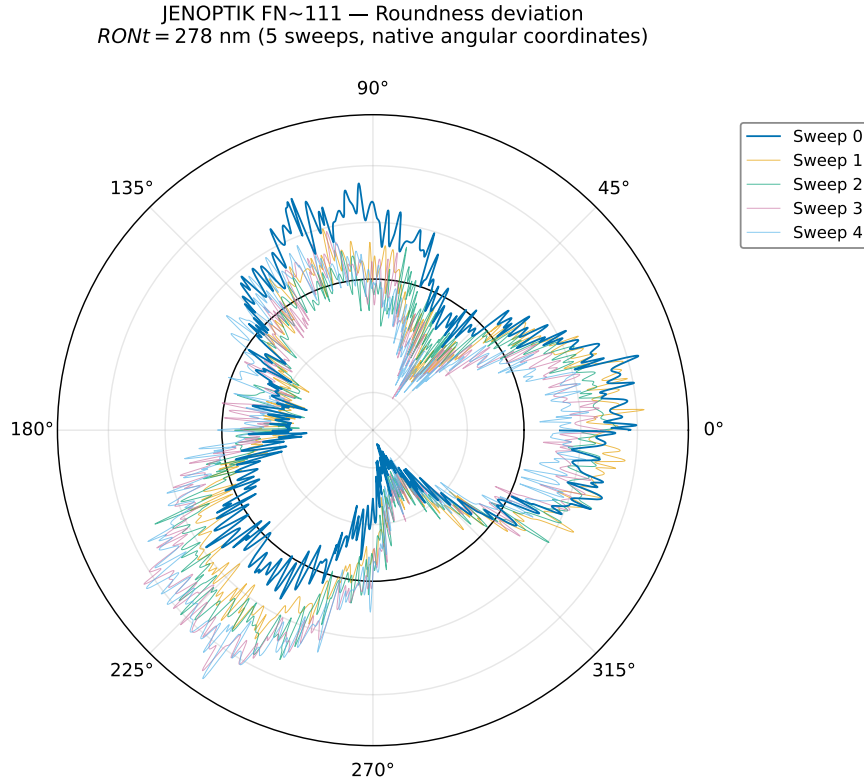


Figure 3: Ceramic roundness in polar coordinates (LSGI + Gaussian low-pass, 50% at
 15 UPR). Five sweeps at native positions. $RONt = 278 \pm 22$ nm.

295 The steel roundness polar representation (FN 101, Flick flat) is provided in

the Supplementary Materials (Fig. S11).

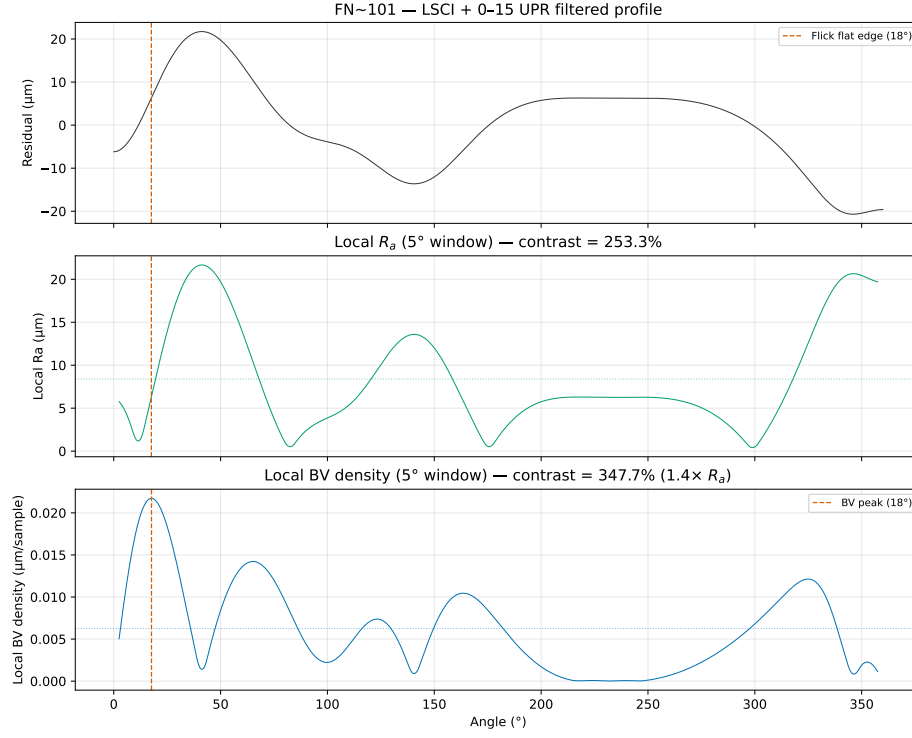


Figure 4: Flick flat detection on the FN 101 magnification standard. BV density (bottom) shows a clear spike at the Flick flat angular position, whereas R_a computed in a sliding window (top) shows no detectable response at the reconstruction resolution. The sliding-window width was matched to the certified angular extent of the Flick flat as specified in the DAkKS-DKD certificate (Art. 532 528); identical window parameters were used for both descriptors.

The Flick flat produces a clear response in BV density while showing no detectable response in a sliding-window R_a analysis at the reconstruction resolution (Fig. 4), illustrating the potential complementary sensitivity of gradient-based descriptors to localised geometric features.

301 5. Discussion

302 5.1. Confirmed Results

303 The analysis rests on a key metrological distinction: an interferogram is
304 an intensity measurement, not a phase map and not a height map. The half-
305 wavelength relation $z = \nu\lambda/2$ becomes valid only after each image has been
306 decoded to a wrapped excess fraction and the ordered excess-fraction sequence
307 has been unwrapped to the continuous fringe-order coordinate ν . This is why
308 the reported comparison uses only reconstructed nanometre profiles, not raw
309 greyscale images and not areal height maps.

310 The most sensitive part of the workflow is the EFM localisation of bright-
311 ring maxima, with an excess-fraction uncertainty of approximately ± 0.05 fringe
312 orders (tightened to ± 0.03 by sinusoidal refinement in 72% of FN 111 frames).
313 A single ϵ_i error of 0.05 fringe corresponds to 13 nm, sub-dominant to the
314 85 nm residual-profile RMS. The chronological unwrapping criterion rejects
315 phase jumps exceeding 0.35 fringe. The median number of detected rings is
316 three for both artefacts within the 240 px central aperture. The five 400° ce-
317 ramic sweeps require a geometric correction: successive useful 360° intervals are
318 shifted by 40° because the physical object angle advances continuously; without
319 this correction, the LSCI fit appears inconsistent even when phase extraction
320 within each sweep is repeatable.

321 The linear drift of $0.0246 \text{ fringe frame}^{-1}$ in the steel unwrapped phase se-
322 quence (versus 0.00021 for the FN 111) is attributed to cumulative bias in the
323 global unwrapping algorithm rather than thermal effects or axial runout [45].
324 The lower fringe contrast of the FN 101 interferograms — likely due to surface
325 finish — reduces per-frame EFM quality; the pipeline applies automatic linear
326 detrending where needed.

327 5.2. Interpretation

328 5.2.1. Eccentricity removal and wobble verification

329 Roundness descriptors are computed after LSCI reference-circle fitting (Sec. 3.3),
330 which removes the first harmonic. A shared model across all five sweeps yields

331 $R^2 = 0.99991$ with per-sweep amplitude variation of 63 nm (0.18%); the manufacturer-
 332 specified wobble ($\pm 12 \mu\text{rad}$) bounds the observed variation an order of magni-
 333 tude below $RONt = 278$ nm. The polar representation (Fig. 3) shows five
 334 overlaid sweeps repeatable to ± 50 nm, illustrating that the residual is domi-
 335 nated by low-order form deviation rather than random roughness.

336 5.3. Comparison with ISO Parameters

337 A central question of this work is whether gradient-based functional descrip-
 338 tors provide additional descriptive sensitivity alongside classical ISO parameters
 339 when applied to the same reconstructed profiles. To answer this rigorously, every
 340 descriptor was computed on the same reconstructed nanometre residual profiles.

341 **Mathematical equivalences.** The L^1 norm is defined as $\frac{1}{N} \sum |p_i - \bar{p}|$,
 342 which is identical to the definition of R_a . The L^2 norm is $\sqrt{\frac{1}{N} \sum (p_i - \bar{p})^2}$,
 343 identical to R_q . Computing both on the same centred residual yields *exact*
 344 *numerical equality*: $R_a - L^1 = 0$ and $R_q - L^2 = 0$ to machine precision for
 345 both artefacts (Table 2). These equalities are not empirical findings; they are
 346 definitional consequences. Labelling the arithmetic mean of absolute deviations
 347 as L^1 rather than R_a provides no new metrological information and should not
 348 be presented as a methodological advance.

349 It is emphasised that neither the L^p norms nor the Sobolev and BV semi-
 350 norms are presented as novel mathematical constructs; they are standard tools
 351 whose definitions are restated for completeness. Their value in the present con-
 352 text is threefold. First, the functional-analytic taxonomy places ISO parameters
 353 at specific positions (L^1 , L^2 , L^∞) within a larger space of possible descriptors,
 354 revealing that descriptors outside the ISO set — BV density, $W^{1,2}$ seminorm
 355 — respond differently to the same perturbations. Second, the synthetic-defect
 356 injection methodology provides a controlled, like-for-like comparison framework
 357 that can be applied to any descriptor computed from the same residual profile;
 358 the framework itself, rather than any individual norm, constitutes the method-
 359 ological contribution. Third, the Flick flat on the certified FN 101 standard pro-
 360 vides a concrete existence proof: a real, DAkkS-DKD-certified artefact whose

defining geometric feature is detected unambiguously by BV density while remaining below the detection threshold of sliding-window R_a (Fig. 4).

Extrema: R_z versus L^∞ . $R_z/L^\infty \approx 2$ for both artefacts, consistent with symmetric positive and negative excursions. L^∞ captures the single largest one-sided excursion; the ratio serves as a simple symmetry indicator.

Variation-based descriptors. A controlled sensitivity analysis (Supplementary Table S2) examined five synthetic defect types injected into the FN 111 residual profile. Amplitude-averaging descriptors (R_a , R_q , R_z) showed no detectable change at the reconstruction resolution — a single 100 nm spike changed R_a by less than 0.05%, below the per-frame phase-extraction precision — while BV density increased by +18% and the $W^{1,2}$ gradient seminorm by +622% for the same spike (Fig. 5). ISO shape parameters responded to asymmetry (R_{sk} changed by −1104% for a step) but not to symmetric localised features. R_{sk} and BV density appear complementary in the tested scenarios: R_{sk} responds to asymmetry, BV density responds to localised gradients.

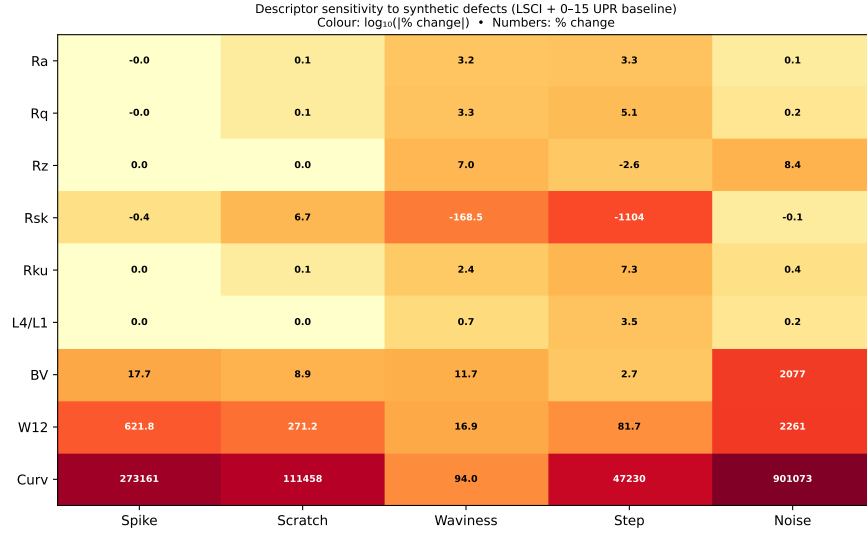


Figure 5: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Percentage change from the unperturbed baseline; bold labels indicate changes exceeding 10%.

Supporting analyses — descriptor correlation matrices, power spectral density comparison, Cohen’s d effect sizes, critical descriptor assessment, and L^p norm convergence — are provided in the Supplementary Materials (Figs. S13–S17). Given the number of descriptors compared (~ 15 across two artefacts), the reported differences should be interpreted as exploratory rather than confirmatory; no formal multiple-comparison correction has been applied.

Practical considerations. A practical suggestion for interferometric profile and roundness metrology is to retain R_a , R_q , R_z , R_{sk} , $RONt$ and $RONq$ as the primary engineering layer compliant with ISO 4287, and to consider supplementing them with BV density (in angular units, nm deg^{-1}) as a gradient-sensitive functional-analysis complement. The $W^{1,2}$ gradient seminorm is correlated with BV density in the present datasets (within the FN 111: Pearson $r = 0.95$, $n = 5$ sweeps; within the FN 101: $r = 0.93$, $n = 5$). The pooled $r = 0.94$ (95% CI $[0.72, 0.99]$, $n = 10$) is reported for descriptive purposes only — the sweeps are not independent observations and this pooled correlation should be interpreted with caution. $W^{1,2}$ may add limited independent information beyond BV density alone given the small sample. The L^4/L^1 peakiness ratio provides a supplementary distribution-shape axis with a direct physical interpretation as the amplification of dominant profile excursions relative to R_a . These additional descriptors are computed from the same residual profile without additional hardware or changes to the EFM pipeline. The cross-instrument validation suggests that gradient-based descriptors are applicable to any 1D profile, whether from interferometry or tactile roundness measurement.

Planned versus exploratory comparisons. The synthetic-defect responses (Supplementary Table S2, Fig. 5) and the Flick flat detection (Fig. 4) were planned comparisons motivated by the study’s objective of assessing gradient-based descriptor sensitivity. The correlation matrices, power spectral density comparison, Cohen’s d effect sizes, L^p norm convergence analysis, and the cross-descriptor stability comparison (Supplementary Figs. S13–S17) are exploratory and are intended to characterise the descriptor space rather than to test specific hypotheses. This distinction should be borne in mind when interpreting the

407 reported associations. The present study is a proof-of-concept demonstration;
 408 generalisability to other artefact types, materials, and measurement instruments
 409 remains to be established.

410 5.4. Limitations

411 Several limitations should be acknowledged. First, the uncertainty analysis
 412 is restricted to within-run Type A repeatability from five sweeps in a single
 413 continuous rotation. The small $n = 5$ limits statistical power (relative dif-
 414 ferences below $\sim 30\%$ cannot be resolved), and between-run reproducibility
 415 (independent setups, thermal cycles, re-centring) has not been assessed. A full
 416 GUM-compliant uncertainty budget, incorporating type-B evaluations of laser
 417 wavelength stability, refractive-index variations, Abbe error, camera nonlinear-
 418 ity, and form-removal uncertainty, is beyond the scope of this study; the reported
 419 expanded uncertainties should be interpreted as lower bounds.

420 Second, the validation is restricted to two artefacts from a single manufac-
 421 turer (JENOPTIK), both ~ 30 mm diameter; generalisability to other sizes,
 422 materials, or surface finishes has not been tested. Third, the synthetic-defect
 423 analysis uses idealised mathematical defects (isolated spike, uniform scratch,
 424 pure sinusoid) injected into the FN 111 ceramic profile only; the response to real
 425 manufacturing defects with complex morphologies may differ, and the defect-
 426 to-background ratio for the FN 101 is approximately 500 times smaller, so the
 427 detection sensitivity reported here may not generalise to artefacts with substan-
 428 tially larger form deviations. Fourth, BV density and $W^{1,2}$ seminorm carry
 429 intrinsic sampling-interval dependence requiring conversion to angular units
 430 for cross-instrument comparison. Fifth, the Flick flat result (Supplementary
 431 Fig. S18) is demonstrated on a single artefact with a known macroscopic feature;
 432 performance for subtler, micrometre-scale defects has not been characterised.

433 5.5. Metrological Implications

434 The results carry several implications for profile and roundness metrology
 435 practice. The finding that BV density responds to localised gradients that am-
 436 plitude averaging may not capture suggests — subject to further validation —

437 a potential role for gradient-based descriptors in applications where isolated
438 defects are of concern; possible examples include quality control of bearing sur-
439 faces, optical components, or precision seals. Because these descriptors are
440 computed from the same residual profile used for ISO 4287 parameters, their
441 adoption does not require changes to existing measurement acquisition proto-
442 cols.

443 The ISO 4287 framework remains the appropriate baseline for surface tex-
444 ture specification and conformity assessment. The gradient-based descriptors
445 discussed here are proposed as optional supplementary indicators, not as re-
446 placements for standardised parameters. Their metrological utility would be
447 strengthened by inter-laboratory comparisons, by evaluation on a wider range
448 of artefact types, and by the development of agreed procedures for converting
449 sampling-dependent units to instrument-independent angular gradient units.
450 Until such steps are taken, BV density and related descriptors should be re-
451 garded as research-grade tools whose diagnostic value has been observed under
452 controlled laboratory conditions on two specific artefacts.

453 5.6. Future Work

454 The present study suggests several directions for further investigation. (i) A
455 between-run reproducibility study with independent setups and re-centring would
456 establish whether the gradient-based descriptors retain their relative stability
457 under realistic laboratory conditions. (ii) Extending the validation to artefacts
458 with different diameters, materials (e.g., glass, carbide), and surface finishes
459 would test the generalisability of the findings. (iii) A systematic characterisation
460 of BV density response to real manufacturing defects (scratches, pits, inclusions)
461 of controlled dimensions would bridge the gap between the present synthetic-
462 defect analysis and industrial inspection scenarios. (iv) An inter-laboratory
463 comparison using a common artefact set and a standardised processing pipeline
464 would provide the reproducibility data needed to assess whether BV density
465 could eventually be considered for standardisation. (v) The development of
466 agreed angular-unit conventions for sampling-dependent gradient descriptors

would facilitate cross-instrument comparison and potential inclusion in reference software for roundness evaluation.

Data Availability

The analysis code, processed results, and all figures are available at <https://github.com/dawidkucharski/banach-profile-metrology> (release v1.0.0). The raw interferometric PNG archives (~32 GB) are available from the corresponding author upon reasonable request.

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